Project Report: Automated Podcast Transcription and Topic Segmentation

Project Title: Automated Podcast Transcription and Topic Segmentation

Dataset Used: TED Talks

Technology Stack: Python, OpenAI Whisper, Streamlit, NLTK, Plotly, Transformers

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1. Introduction

In recent years, podcasts and long-form audio have become a primary source of information for education, news, and entertainment. These audio files often cover diverse subjects, including technology, science, and culture. However, most podcasts are long in duration, making it difficult for listeners to quickly identify relevant topics or retrieve specific information. Listening to an entire episode to find a specific discussion is often time-consuming and inefficient.

To overcome this problem, there is a need for an intelligent system that can automatically transcribe podcasts, divide them into meaningful topic segments, and provide summaries, keywords, and sentiment insights. The **Automated Podcast Transcription and Topic Segmentation** project addresses this challenge by using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to convert unstructured audio into structured, searchable, and easy-to-understand content.

2. Problem Statement

Long-form audio content contains valuable information, but its duration makes it difficult for users to locate specific topics quickly. There is currently no efficient mechanism to navigate audio files based on the specific sub-topics or themes discussed within them. Therefore, the problem is to design a system that can:

- Automatically convert podcast audio into accurate text.
- Identify logical topic boundaries within the transcript.
- Generate concise summaries, key terms, and sentiment analysis for each section.
- Provide a user-friendly interface for semantic navigation and analysis.

3. Objectives of the Project

The main objectives of this project are:

1. To convert raw podcast audio into accurate text using advanced speech-to-text models (OpenAI Whisper).
2. To segment the transcript into meaningful topics using NLP algorithms (TextTiling).
3. To extract important keywords and generate abstractive summaries for each segment.
4. To analyze the sentiment of each topic to gauge the emotional flow of the conversation.

5. To provide an interactive frontend (Streamlit) for viewing transcripts, visualizations, and searching content.

4. Dataset Selection: TED Talks

For the development and validation of this system, we selected the **TED Talks Dataset** as our primary corpus.

- **Linguistic Diversity:** TED speakers come from global backgrounds, allowing us to stress-test the Whisper model's ability to handle various accents and speaking speeds.
- **Semantic Complexity:** The dense, technical nature of the talks (Science, Technology, Design) rigorously tested our Summarization (DistilBART) and Keyword Extraction models.
- **Structure:** The clear narrative structure of TED Talks provided an ideal ground truth for validating our segmentation logic.

5. Scope of the Project

The scope of this project includes:

- Processing raw audio files (MP3/WAV) to remove noise and normalize formats.
- Automatic transcription using the OpenAI Whisper ASR model.
- Topic segmentation using the TextTiling NLP technique.
- Keyword extraction using NLTK and Summarization using DistilBART.
- Visualization through a web-based Streamlit dashboard.

The project does not include live streaming audio or real-time transcription.

6. System Architecture

The system follows a modular, pipeline-based architecture:

Audio Input → **Preprocessing** (Cleaning) → **ASR** (Whisper) → **NLP** (TextTiling & NLTK) → **Intelligence** (Summarization & Sentiment) → **Frontend** (Streamlit Dashboard)

7. Module Description

7.1 Audio Preprocessing Module

This module prepares raw audio for transcription. It utilizes soundfile and numpy to perform audio normalization and noise handling to improve transcription accuracy.

7.2 Transcription Module

The transcription module converts audio into text using the **OpenAI Whisper** model (supporting tiny, base, and small architectures). It generates precise timestamps for every word to align text with audio.

7.3 Topic Segmentation Module

This module identifies topic changes within the transcript using the **TextTiling** algorithm. It detects shifts in vocabulary usage to mark where one subject ends and another begins, ensuring each segment represents a meaningful discussion.

7.4 Intelligence Module (Keywords, Summaries, Sentiment)

- **Keywords:** Important terms are extracted using **NLTK** tokenization to populate the bubble chart.
- **Summarization:** The **DistilBART** Transformer model generates a 2-sentence abstract for each segment.
- **Sentiment: TextBlob** performs segment-level sentiment analysis to calculate polarity scores.

7.5 Frontend Visualization Module

The Streamlit frontend displays:

- Full interactive transcript.
- **Packed Bubble Chart:** A custom "collision-free" visualization of top keywords.
- **Sentiment Timeline:** A Gantt chart showing the conversation flow.
- **Deep Search:** Users can navigate topics using timestamps and play specific audio slices.

8. Technology Stack

Backend Technologies:

- Python 3.9+
- OpenAI Whisper (ASR)
- SoundFile, NumPy (Audio Processing)

NLP and Machine Learning:

- NLTK (Natural Language Toolkit)
- HuggingFace Transformers (DistilBART)
- TextBlob (Sentiment)

Frontend Technologies:

- Streamlit (Web Framework)
- Plotly Express (Interactive Charts)
- Custom CSS (Styling)

9. Project Structure

The project is organized into backend and frontend modules:

- **src/** contains preprocessing, transcription, segmentation, and keyword logic.
- **ui_app.py** acts as the main entry point for the Streamlit dashboard.
- **.gitignore** manages version control by excluding heavy media files.

10. Implementation Journey (Week-by-Week)

Week 1: Project Setup & Feasibility

- Selected TED Talks as the target dataset.
- Initialized the Python environment (.venv) and Git repository.
- Configured .gitignore to whitelist the src/ folder while excluding heavy media files.

Week 2: The Transcription Pipeline

- Implemented clean_audio() to normalize audio formats.
- Integrated **OpenAI Whisper** and wrote the transcribe_audio() script.
- Result: Successfully converted a 10-minute audio file into raw text.

Week 3: Topic Segmentation Logic

- Implemented the **TextTiling algorithm**.
- Logic: The system scans for shifts in vocabulary (e.g., a switch from "economics" words to "climate" words) and marks a segment break at that specific timestamp.

Week 4: Summarization & Sentiment Analysis

- Integrated **DistilBART** to generate 2-sentence abstracts for every topic block.
- Applied **TextBlob** to assign sentiment scores (Positive/Negative) to each segment.

Week 5: Full-Stack UI Integration

- Built the frontend using **Streamlit**.
- Developed the **Interactive Timeline** (Gantt chart) using Plotly Express.
- Implemented **Semantic Search**, allowing users to filter content by keywords and play specific audio slices.

Week 6: System Optimization & Final Polish

- **Performance:** Implemented Streamlit Caching (@st.cache_resource) to load heavy AI models only once.
- **Visualization Upgrade:** Replaced the standard Word Cloud with a custom **Packed Bubble Chart**.
- **Styling:** Applied custom CSS for a professional "Dodger Blue" theme.

11. Future Enhancements

- Speaker diarization (identifying "Speaker A" vs "Speaker B").

- Semantic search across multiple podcast files.
- Multi-language transcription support.
- Export functionality to download summaries as PDF reports.

12. Conclusion

The **Automated Podcast Transcription and Topic Segmentation** project provides an efficient solution for navigating long audio content. By combining AI, NLP, and advanced visualization, the system makes podcast content more accessible, searchable, and informative. The successful integration of TED Talks data demonstrates the system's capability to handle complex, intellectual discourse with high accuracy.